

Weak Learning Failure Alone Is Not Gradient Starvation: Matched Counterfactuals and Exact Finite-Width Dynamics

Anonymous authors

Working draft

Abstract

A weak feature can remain poorly learned because its weak-only task is slow, a stronger cue changes cross-entropy weighting, or response-specific parameter geometry suppresses its update. These explanations are not causally equivalent. We introduce a matched weak-only counterfactual: the pair shares initialization, data, weak inputs, and noise; only the strong input is ablated. Causal starvation requires both outcome suppression and a nondegenerate weak-only first hit; gate failure instead leaves causal status indeterminate.

For full-batch logistic gradient flow, we derive exact finite-width dynamics, $\dot{M}_w = c_w^\top q(r)/n$ and $\dot{r} = \Theta q(r)/n$, which separate sample weighting from response geometry without a mean-field closure. The pathwise identity $\Delta = P - N$ determines when later suppression outweighs earlier transfer. Controlled studies show starvation exists but depends on lag, architecture, initialization, target, and horizon: some regimes yield causal certificates; others show only crossings or fail the weak-only gate. A frozen-tangent surrogate predicts held-out crossing events but not calibrated trajectories or causal status. Initialization and local-tube results provide conditional prediction; Counterfactual Drift Correction gives a unique minimum-norm local intervention in a fixed parameter chart. Neither result supplies a global positive-lag predictor or a broadly superior optimizer.

1 Introduction

When a model learns one predictive feature and neglects another, the neglected feature is often called “starved.” But a small weak-feature response is not by itself causal evidence: the weak signal may simply be too difficult to learn under the chosen target and horizon. The distinction matters especially in recurrent networks, where cue strength, temporal separation, shared parameters, and evolving geometry jointly determine learning.

We isolate the effect of the strong cue with a matched intervention. A both-feature model is paired with a weak-only model that shares initialization, labels, samples, weak inputs, and noise; only the strong input channel is ablated. For their weak responses, let $\Delta = M_w^B - M_w^W$ and $d = \Delta'$. Positive d is relative transfer, negative d is rate suppression, and negative Δ is outcome suppression. We call the outcome *causal starvation* only when the weak-only trajectory also reaches a prespecified target at a nondegenerate positive time. If that gate fails, causal status is indeterminate: neither starvation nor its absence has been established.

Four findings.

1. **A matched weak-only counterfactual is necessary for causal attribution.** It separates a feature that would not learn on its own from one whose learning is suppressed by the strong cue. It also separates transient rate suppression from outcome suppression: later negative drift matters only if it repays earlier positive transfer.
2. **Exact finite-width dynamics separate cross-entropy weighting from response geometry.** For signed sample logits r under full-batch logistic gradient flow,

$$\dot{M}_w = \frac{1}{n} c_w^\top q(r), \quad \dot{r} = \frac{1}{n} \Theta q(r), \quad (1)$$

where $q_i(r) = \sigma(-r_i)$, $\Theta = JJ^\top$, and $c_w = J\nabla M_w$. These are finite-width chain-rule identities, not a mean-field closure. For the paired response, the exact pathwise balance $\Delta = P - N$ shows that outcome

suppression occurs precisely when accumulated negative relative drift exceeds accumulated positive relative drift.

3. **Starvation exists, but is regime-dependent rather than universal.** In the controlled linear grid, every measured cell at lags 0, 2, and 4 has a positive weak-AUC deficit and a learnable weak-only control, whereas no lag-8 weak-only trajectory reaches its target. At one nonlinear cell, tanh yields complete causal certificates in 3/8 seeds; GRU exhibits drift and response crossings in 5/8 seeds but no causal certificate because its weak-only gate fails. Semi-real generated-cue and threshold-robust extensions are likewise negative or indeterminate. These fixed-grid counts establish neither universal prevalence nor an architecture ranking.
4. **Prediction and intervention have precise local or conditional limits.** Realized initialization gives an exact local sign in the dense-linear setting, and stronger crossing guarantees require independently established parameter-tube or kernel-movement bounds. A frozen-tangent model classifies held-out crossing events, but absent those bounds it is an empirical event heuristic rather than calibrated dynamics or causal prediction. Counterfactual Drift Correction (CDC) is the unique minimum-Euclidean-norm feasible uncapped target-attaining correction in a fixed parameter chart, with only a local one-step guarantee. Its comparisons support neither canonical parity nor broad optimizer superiority.

The resulting contribution is a disciplined chain from exact dynamics to matched causal intervention, pathwise attribution, and local mechanism control. We do not solve a general positive-lag optimization-time mean-field theory, derive a universal boundary in cue strength and lag, or claim that all weak learning failure is starvation. Instead, we give exact identities, assumption-scoped predictions, and regime-specific evidence with explicit causal gates.

2 Related work

Feature starvation and shortcut learning. Cross-entropy can favor one predictive feature strongly enough to starve another, and Spectral Decoupling was introduced to counter that mechanism [Pezeshki et al., 2021]. Shortcut learning and simplicity bias describe the broader tendency to exploit predictive but unintended cues [Geirhos et al., 2020, Shah et al., 2020]. Our contribution is diagnostic: the matched weak-only trajectory tests whether the neglected feature would have learned without the strong cue, while direct drift and pathwise balance distinguish rate suppression from an adverse outcome.

Methods such as Group DRO, Just Train Twice, and last-layer retraining target worst-group performance under spurious correlations [Sagawa et al., 2020, Liu et al., 2021, Kirichenko et al., 2023]. We address a different question—causal attribution of feature-level learning dynamics—and do not propose a competing robustness benchmark. Our generated-cue image study preserves real core pixels but is not an unmodified natural-spurious-correlation setting.

Wide and recurrent learning dynamics. Neural tangent kernels characterize infinite-width gradient dynamics [Jacot et al., 2018], and wide-network linearization motivates frozen-tangent approximations [Lee et al., 2019]. Our response-logit equations are instead exact at finite width. Freezing their kernels defines a nonlinear logistic tangent model, but transferring its predictions to a trained network requires kernel-movement control that we do not establish for tanh RNNs or GRUs. Low-dimensional overlap theories for low-rank RNNs show that loss-invisible modes can retain training history [Ger and Barak, 2026], cautioning against closure in only the visible response coordinates. Mean-field descriptions of structure and disorder in trained recurrent circuits [Clark et al., 2026] study a distinct long-time equilibrium setting. We therefore do not claim a general recurrent optimization-time mean-field theory.

Gradient surgery. PCGrad projects conflicting task gradients [Yu et al., 2020]; Bloop stabilizes an auxiliary projection with an exponential moving average [Hsieh et al., 2024]. CDC also uses a projection, but its protected direction is the gradient of a strong-feature response and its target comes from the matched weak-only trajectory. That extra counterfactual requires a constructible ablation, a second model, lockstep training, and full-batch gradients. Our comparisons use adapted shadow-target variants and support neither equivalence to the original methods nor broad CDC superiority.

3 Paired causal diagnosis and definitions

3.1 Paired logistic gradient flow

For condition $C \in \{B, W\}$, let $\theta_C(\tau) \in \mathbb{R}^P$ follow full-batch gradient flow on signed logits $r_i^C(\theta) = y_i f_C(x_i; \theta)$:

$$L_C(\theta) = \frac{1}{n_C} \sum_{i=1}^{n_C} \log(1 + e^{-r_i^C(\theta)}), \quad \dot{\theta}_C = -\nabla L_C(\theta_C). \quad (2)$$

The two conditions share initialization and all data except that the strong channel is zeroed under W. This matched weak-only path tests whether the weak signal can meet the specified target without the strong cue; it is the causal identification premise that distinguishes suppression from an unsupported weak- learning label. Let M_s^C, M_w^C be common differentiable strong and weak response functionals. Shared initialization gives

$$\Delta(0) = 0, \quad \Delta(\tau) = M_w^B(\tau) - M_w^W(\tau). \quad (3)$$

Optimization time τ is distinct from sequence index k .

The controlled dense linear RNN is

$$h_{k+1} = Wh_k + Bx_k, \quad f(x) = c^\top h_T, \quad (4)$$

with $W_{ij}(0) \sim \mathcal{N}(0, g^2/N)$, $B_{ia}(0) \sim \mathcal{N}(0, 1/N)$, and $c_i(0) \sim \mathcal{N}(0, 1/N)$. The width scaling of B is essential for a non-divergent projected metric. We additionally study tanh RNNs and GRUs. For nonlinear models the two responses are common odd symmetric unit-probe contrasts, $[f(x_a^+) - f(x_a^-)]/2$, not assumed additive coordinates of every data logit.

3.2 Weak learning failure, causal labels, and metrics

Let

$$d(\tau) = \Delta'(\tau) = F_w^B(\tau) - F_w^W(\tau). \quad (5)$$

We call $d > 0$ relative transfer and $d < 0$ relative rate suppression. We call $\Delta < 0$ *outcome suppression*, whether it follows a positive excursion or is present from the start of optimization. A response crossing is the special case in which Δ returns to zero after a positive excursion and then becomes negative. Weak-only nonlearnability under the specified gate means that the weak-only trajectory does not first hit the prespecified target β within horizon H . We reserve *causal starvation* for outcome suppression when that weak-only trajectory does reach the target at a nondegenerate positive time. Thus weak learning failure alone is not gradient starvation. If the weak-only model does not reach β , an outcome crossing is descriptive but not a causal- starvation certificate; the failed gate also is not evidence that starvation is absent. Causal status is indeterminate under the specified target and horizon.

We report first-hitting-time delay when both hits are observed. Formally, $\tau_W(\beta; H) = \inf\{\tau \in [0, H] : M_w^W(\tau) \geq \beta\}$, with $\inf \emptyset = \infty$; the learnability gate requires $\tau_W(\beta; H) \in (0, H]$, excluding targets met at initialization. The horizon-wide weak AUC gap is

$$\mathcal{A}_w = \int_0^H (M_w^W(\tau) - M_w^B(\tau)) \, d\tau, \quad (6)$$

computed by the logged-grid trapezoid rule. To expose rather than hide sensitivity to the single prespecified β , Section 9 also reports the threshold-free terminal response $B_W(H) = M_w^W(H)$ across seeds and its post-hoc threshold profile. Terminal thresholding is not generally equivalent to the prespecified first-hit event on a nonmonotone trajectory, so the profile is a sensitivity diagnostic rather than a replacement gate. GSI-5 remains a margin- weight diagnostic, not a definition of starvation.

4 Exact finite-width dynamics for the paired diagnosis

The mechanism-level decomposition is exact before any width limit: with $q_i(r) = \sigma(-r_i)$ and response-specific geometry $c_w = J\nabla M_w$, the weak response obeys $\dot{M}_w = c_w^\top q(r)/n$. The vector $q(r)$ records cross-entropy

weighting across signed sample logits, while c_w records how the chosen weak response couples to those logits through parameter space. No DMFT or two-response closure is needed for this identity; such closures arise only in stronger predictive approximations.

Proposition 1 (Universal response-logit cross-kernel flow). *At every finite width and every optimization time at which M_a^C is differentiable,*

$$\dot{M}_a^C = \frac{1}{n_C} \sum_i K_{ai}^C \sigma(-r_i^C), \quad K_{ai}^C = \langle \nabla M_a^C, \nabla r_i^C \rangle. \quad (7)$$

No width limit or two-mode realization is required. We label this elementary chain-rule identity as a proposition: its importance here is the exact finite-width scope and the causal separation it enables, not proof difficulty.

Corollary 2 (Closed projected flow under exact realization). *If the signed logits admit the exact parameter-independent representation $r_i^C = \sum_b z_{ib}^C M_b^C$, then*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C, \quad G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle, \quad g_b^C = \frac{1}{n_C} \sum_i z_{ib}^C \sigma(-z_i^C \cdot M^C). \quad (8)$$

If $r_i^C = z_i^C \cdot M^C + r_i^\perp$, the exact equation instead contains

$$R_a^C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle, \quad \dot{M}_a^C = (G^C g^C)_a + R_a^C. \quad (9)$$

Corollary 3 (Exact full empirical-NTK dynamics). *Let $J^C(\tau) \in \mathbb{R}^{n_C \times P}$ have row $\nabla r_i^C(\theta_C(\tau))^\top$, and define*

$$\Theta^C(\tau) = J^C(\tau) J^C(\tau)^\top, \quad c_a^C(\tau) = J^C(\tau) \nabla M_a^C(\theta_C(\tau)). \quad (10)$$

Writing $q(r) = \sigma(-r)$ componentwise, the exact finite-width flow is

$$\dot{r}^C = \frac{1}{n_C} \Theta^C(\tau) q(r^C), \quad \dot{M}_a^C = \frac{1}{n_C} c_a^C(\tau)^\top q(r^C). \quad (11)$$

No width limit or response-coordinate realization is required.

Definition 4 (Initialization-frozen nonlinear logistic surrogate). *For one condition, let $\Theta_0 = \Theta(0)$ and $c_0 = c_a(0)$. Define*

$$\dot{\bar{r}} = \frac{1}{n} \Theta_0 q(\bar{r}), \quad \dot{\bar{M}} = \frac{1}{n} c_0^\top q(\bar{r}), \quad (\bar{r}(0), \bar{M}(0)) = (r(0), M(0)). \quad (12)$$

The gate remains nonlinear. This system is exact for the initialization-linearized tangent model, not for the trained nonlinear network.

Theorem 5 (Conditional continuous and secant-SGD comparison). *Use Euclidean and induced operator norms. Suppose on $[0, t]$ that $\|\Theta(u) - \Theta_0\| \leq B_\Theta$, $\|c(u) - c_0\| \leq B_c$, and both logistic weight vectors have norm at most V . Set $a = \|\Theta_0\|/(4n)$,*

$$\phi_a(t) = \frac{e^{at} - 1}{a}, \quad \psi_a(t) = \frac{e^{at} - 1 - at}{a^2}, \quad (13)$$

with $\phi_0(t) = t$ and $\psi_0(t) = t^2/2$. Then

$$\|r(t) - \bar{r}(t)\| \leq \frac{B_\Theta V}{n} \phi_a(t), \quad (14)$$

$$|M(t) - \bar{M}(t)| \leq \frac{B_c V}{n} t + \frac{\|c_0\| B_\Theta V}{4n^2} \psi_a(t). \quad (15)$$

For explicit full-batch SGD, let $\delta\theta_k = \theta_{k+1} - \theta_k$ and define exact segment kernels

$$\tilde{J}_k = \int_0^1 J(\theta_k + s\delta\theta_k) ds, \quad \tilde{\Theta}_k = \tilde{J}_k J(\theta_k)^\top, \quad (16)$$

$$\tilde{m}_k = \int_0^1 \nabla M(\theta_k + s\delta\theta_k) ds, \quad \tilde{c}_k = J(\theta_k) \tilde{m}_k. \quad (17)$$

The true increments are exactly $r_{k+1} - r_k = \eta \tilde{\Theta}_k q(r_k)/n$ and $M_{k+1} - M_k = \eta \tilde{c}_k^\top q(r_k)/n$. If $\eta > 0$ and, for every $0 \leq j < k$, $\|\tilde{\Theta}_j - \Theta_0\| \leq D_\Theta$, $\|\tilde{c}_j - c_0\| \leq D_c$, and $\|q(r_j)\|, \|q(\bar{r}_j)\| \leq V$, then the explicit-Euler version of Eq. (12) obeys

$$\|r_k - \bar{r}_k\| \leq \frac{D_\Theta V}{n} \Phi_k, \quad (18)$$

$$|M_k - \bar{M}_k| \leq \frac{k\eta D_c V}{n} + \frac{\eta \|c_0\| D_\Theta V}{4n^2} \Psi_k, \quad (19)$$

where

$$\Phi_k = \frac{(1 + \eta a)^k - 1}{a}, \quad \Psi_k = \frac{(1 + \eta a)^k - 1}{\eta a^2} - \frac{k}{a}, \quad (20)$$

with limits $\Phi_k = k\eta$ and $\Psi_k = \eta k(k-1)/2$ at $a = 0$. Apply the bounds separately to B, W and add them to bound the paired response gap. Frozen signs or brackets transfer only when their strict margins exceed the corresponding summed errors; first hitting times additionally require Corollary 14's prehistory, transversality, radius, and mesh hypotheses.

Theorem 5 is conditional: this paper does not prove the required instantaneous or secant-kernel movement bounds for tanh or GRU. A held-out event-classification result cannot substitute for that premise. In particular, the theorem is finite-width and is not a DMFT closure.

For channel-localized modes of the dense linear RNN, differentiation across the disjoint (c, B, W) blocks gives the exact finite-width geometry

$$G_{ab} = p_a^\top p_b + \mathbf{1}\{\text{ch}(a) = \text{ch}(b)\} q_a^\top q_b + \langle S_a, S_b \rangle_F, \quad (21)$$

where $p_a = W^{\ell_a} b_a$, $q_a = (W^{\ell_a})^\top c$, and

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (22)$$

Theorem 6 (Exact realized-initialization rate-gap criterion). *In the noiseless positive rank-one dense-linear pair, let the strong cue have strength $\rho > 0$ at lag zero, the unit weak cue have lag ℓ , and define*

$$K = G_{ws}(0) = b_s^\top W^\ell b_w, \quad (23)$$

$$A = G_{ww}(0) = \|W^\ell b_w\|^2 + \|(W^\ell)^\top c\|^2 + \|S_w\|_F^2, \quad (24)$$

$$S_w = \sum_{j=0}^{\ell-1} (W^j)^\top c (W^{\ell-1-j} b_w)^\top, \quad (25)$$

$$s_B = \sigma[-(\rho M_s(0) + M_w(0))], \quad s_W = \sigma[-M_w(0)]. \quad (26)$$

(The recurrent sum is zero for $\ell = 0$.) The exact initial equal-time weak-rate gap is

$$d_0 = s_B(\rho K + A) - s_W A. \quad (27)$$

Consequently, initial transfer, suppression, and the boundary occur respectively when

$$K \geq \frac{A}{\rho} \left(\frac{s_W}{s_B} - 1 \right) \quad \text{or equality holds.} \quad (28)$$

This is an exact pre-trajectory criterion from the realized initialization. Since shared initialization gives $\Delta(0) = 0$ and $\Delta'(0) = d_0$, it also implies

$$d_0 < 0 \implies \Delta(t) < 0, \quad d_0 > 0 \implies \Delta(t) > 0 \quad \text{for all sufficiently small } t > 0. \quad (29)$$

These strict local outcome signs are existential and provide no quantitative horizon. At the boundary $d_0 = 0$, first order is silent and licenses neither sign. The criterion is not a distribution-only positive-lag phase boundary: under the stated iid scaling, K and $M_s(0)$ contribute a seed-dependent leading sign.

Theorem 6 is a partial predictor: realized $d(0)$ gives an exact local sign in its stated dense-linear scope, not a global trajectory or a parameter-only law in ρ and lag. Where a higher local jet is proved—as in the zero-lag result with $d(0) = 0$ and $d'(0) < 0$ —it likewise supplies a rigorous local sign only in that stated scope. Stronger positive-lag trajectory prediction requires independently established product-ball or kernel-movement control, for which this paper supplies no nonvacuous general constants.

The proofs are in Appendix A. Equation (8) is exact for the zero-background-noise linear synthetic model. For tanh/GRU probes, we use Proposition 1’s direct autograd drift for inference and report R ; projected and matched-state decompositions are diagnostics only.

5 Separating rate suppression, outcome suppression, and causal starvation

Proposition 7 (Cumulative drift balance). *Let Δ be absolutely continuous on $[0, H]$, with $\Delta(0) = 0$ and $d = \Delta'$ almost everywhere. Write*

$$d_+(t) = [d(t)]_+, \quad d_-(t) = [-d(t)]_+, \quad P(t) = \int_0^t d_+(u) du, \quad N(t) = \int_0^t d_-(u) du. \quad (30)$$

Then $\Delta(t) = P(t) - N(t)$ for every $t \in [0, H]$. Thus strict outcome suppression occurs by H if and only if $N(t) > P(t)$ for some $t \in (0, H]$; this statement permits arbitrary sign changes and suppression from initialization. This is deliberately a bookkeeping proposition: its contribution is the causal separation it enforces, not proof difficulty.

Corollary 8 (Single-crossing tail-area criterion). *Suppose additionally that there is $\tau_d \in (0, H)$ such that $d > 0$ almost everywhere on $(0, \tau_d)$ and $d < 0$ almost everywhere on (τ_d, H) . Define*

$$A_+ = P(\tau_d) = \Delta(\tau_d) > 0. \quad (31)$$

Then $P(t) = A_+$ and $\Delta(t) = A_+ - N(t)$ for $t \geq \tau_d$. Hence a later equality is reached by H if and only if $N(H) \geq A_+$, and strict outcome suppression occurs by H if and only if $N(H) > A_+$. If $d < 0$ on each nontrivial post-crossing interval, the later equality is unique whenever it exists.

$A \rightarrow -$ drift crossing therefore locates a local maximum of Δ ; it does not force Δ to return to zero. Pathwise attribution requires the negative area to repay all positive area already accumulated: outcome suppression begins only when accumulated suppression overtakes accumulated help. Causal *starvation* further requires the independent weak-only learnability gate. A failed gate blocks that certificate but does not establish that starvation is absent.

Corollary 9 (Response-crossing bounds). *Under Corollary 8, if $d(t) \leq -\kappa < 0$ on $[t_0, H]$ with $t_0 > \tau_d$, then, when the right side lies within the horizon,*

$$\tau_{\text{response}} \leq t_0 + \frac{[A_+ - N(t_0)]_+}{\kappa} \leq t_0 + \frac{A_+}{\kappa}. \quad (32)$$

If $d(\tau_d) = 0$ and $d'(t) \leq -\lambda < 0$ on $[\tau_d, \tau_d + r]$, then $N(t) \geq \lambda(t - \tau_d)^2/2$; when $A_+ \leq \lambda r^2/2$,

$$\tau_{\text{response}} \leq \tau_d + \sqrt{2A_+/\lambda}. \quad (33)$$

Theorem 10 (Conditional parameter-tube certificate). *Let $X = (\theta_B, \theta_W)$, let V be the joint paired gradient-flow field, and define $d = L_V \Delta$ and $j = L_V d$. On the radius- R product-norm ball U_R about the shared initialization, suppose uniformly that*

$$\|V(X)\| \leq S_R, \quad -\Lambda \leq j(X) \leq -\lambda < 0, \quad 0 < \lambda \leq \Lambda, \quad (34)$$

with $S_R > 0$, and set $h_R = R/S_R$. If $d_0 > 0$ and $h_R > d_0/\lambda$, there is exactly one transfer-to-rate-suppression crossing before ball exit and

$$\frac{d_0}{\Lambda} \leq \tau_d \leq \frac{d_0}{\lambda}. \quad (35)$$

If $h_R > 2d_0/\lambda$, then $\Delta(t) < 0$ for every $2d_0/\lambda < t \leq h_R$. If also the weak-only drift obeys $\dot{M}_w^W \geq \nu > 0$ uniformly, the target is nondegenerate, and

$$\beta > M_w^W(0), \quad \frac{\beta - M_w^W(0)}{\nu} < h_R, \quad (36)$$

then the weak-only member learns by reaching β at a strictly positive time within the same certified horizon, so the outcome suppression is causally gated starvation rather than suppression alone. If $\beta \leq M_w^W(0)$, the target was met at initialization; this is degenerate attainment, not learnability, and cannot certify causal starvation.

Corollary 11 (Certified finite-horizon safe transfer). *On the same ball, if instead $|j(X)| \leq J$ and $d_0 - Jh_R > 0$, then for $0 < t \leq h_R$,*

$$d(t) \geq d_0 - Jt > 0, \quad \Delta(t) \geq d_0 t - \frac{Jt^2}{2} > 0. \quad (37)$$

Thus suppression is excluded on this certified finite horizon, not for all future optimization time. The constants $S_R, \lambda, \Lambda, \nu$, and J must be established independently rather than inferred from sampled trajectories. No general nonvacuous positive-lag construction is known.

5.1 An ordering-invariant rank-one criterion

Assume deterministic positive cues $z^B = (\rho, 1)$, $z^W = (0, 1)$, $\rho > 0$, and exact two-mode realization. Define

$$u_B = \rho M_s^B + M_w^B, \quad s_B = \sigma(-u_B), \quad (38)$$

$$u_W = M_w^W, \quad s_W = \sigma(-u_W), \quad (39)$$

$$H_B = \rho G_{ws}^B + G_{ww}^B, \quad H_W = G_{ww}^W, \quad (40)$$

$$Q_B = \rho^2 G_{ss}^B + 2\rho G_{sw}^B + G_{ww}^B \geq 0. \quad (41)$$

Theorem 12 (Log drift-ratio identity). *The weak drifts factor as $F_w^B = s_B H_B$ and $F_w^W = s_W H_W$. Where $H_B, H_W > 0$, define*

$$\Psi = \log \frac{F_w^B}{F_w^W} = \log \frac{s_B}{s_W} + \log \frac{H_B}{H_W}. \quad (42)$$

Then $\text{sign}(d) = \text{sign}(\Psi)$ and

$$\Psi' = -s_B(1 - s_B)Q_B + s_W(1 - s_W)H_W + \left(\log \frac{H_B}{H_W} \right)'. \quad (43)$$

Corollary 13 (Unique rate crossing). *Suppose on $[t_0, H]$ that $H_B, H_W > 0$, $\Psi(t_0) > 0$, $\Psi'(t) \leq -\kappa < 0$, and $H - t_0 \geq \Psi(t_0)/\kappa$. Then exactly one rate crossing occurs in $(t_0, H]$ and*

$$\tau_d \leq t_0 + \Psi(t_0)/\kappa. \quad (44)$$

A sufficient primitive condition is

$$s_B(1 - s_B)Q_B \geq a, \quad s_W(1 - s_W)H_W \leq b, \quad (\log(H_B/H_W))' \leq c, \quad a - b - c \geq \kappa > 0. \quad (45)$$

The bounds must be established independently to be predictive. Evaluating Ψ on a completed trajectory is certification, not prediction. Noisy cues generally break the rank-one factorization, and even a rate crossing still requires Corollary 8 for an outcome crossing.

5.2 Hitting-time stability

Corollary 14 (Transverse first-hit stability). *Let continuous $x_N, x : [0, H] \rightarrow \mathbb{R}$ satisfy $\|x_N - x\|_\infty \leq \epsilon$, and define*

$$\tau = \inf\{t \in [0, H] : x(t) \geq \beta\}, \quad \tau_N = \inf\{t \in [0, H] : x_N(t) \geq \beta\}, \quad \inf \emptyset = \infty. \quad (46)$$

Assume x is absolutely continuous, $x(0) < \beta$, and $\tau \in (0, H)$. Choose $0 < r \leq \min\{\tau, H - \tau\}$ such that $x'(t) \geq \kappa > 0$ almost everywhere on $[\tau - r, \tau + r]$, and let

$$\eta = \beta - \max_{0 \leq t \leq \tau - r} x(t) > 0. \quad (47)$$

If $\epsilon < \eta$ and $\epsilon/\kappa \leq r$, then $\tau_N < \infty$ and

$$|\tau_N - \tau| \leq \epsilon/\kappa. \quad (48)$$

For an ordered grid $0 = t_0 < \dots < t_m = H$ of maximum mesh $h = \max_j (t_j - t_{j-1})$, let τ_N^{samp} be the first sampled time with $x_N(t_j) \geq \beta$, and let τ_N^{int} be the first hit of the piecewise-linear interpolant through the samples. If $\epsilon/\kappa + h \leq r$, both are finite and

$$|\tau_N^{\text{samp}} - \tau| \leq \epsilon/\kappa + h, \quad |\tau_N^{\text{int}} - \tau| \leq \epsilon/\kappa + h. \quad (49)$$

For a paired delay, if both reference hits are finite and each trajectory satisfies its own assumptions, then

$$|(\tau_{N,B} - \tau_{N,W}) - (\tau_B - \tau_W)| \leq \epsilon_B/\kappa_B + \epsilon_W/\kappa_W + h_B + h_W. \quad (50)$$

If either hit is infinite, the delay is censored; a common mesh h gives the special case with $2h$.

Descending crossings follow by sign reversal and localization after a strict excursion. Since the response gap has $\Delta(0) = 0$, its later downward outcome crossover must be localized to $[a, H]$ with $\Delta(a) > 0$ and treated using $x = -\Delta$ and target zero; the hit at time zero is not that later crossover. This corollary transfers an independently proved uniform trajectory limit. It does not prove that limit, first-entry semantics, the prehistory margin, or the derivative bound. Thus Theorem 5's summed paired response bound licenses a crossing-time statement only after all of these separate guards hold; an error formula or empirical event match alone is not a hitting-time certificate.

6 A solvable recurrence-invisible zero-lag wide limit

Set lag separation zero and background input/noise zero in the dense linear model; let the realized recurrent matrix W be arbitrary and finite. Both cues then arrive at the final step. Starting from $h_0 = 0$, induction gives $h_t = 0$ for every state entering the final update and hence $h_T = Wh_{T-1} + Bx_{T-1} = Bx_{T-1}$. Thus W is loss-invisible, $\nabla_W L = 0$, and it remains fixed. Writing the input columns as b_s, b_w gives $M_a = c \cdot b_a$. Let $u_{ab} = b_a \cdot b_b$ and $w = c \cdot c$.

Theorem 15 (Every-width six-scalar closure). *At every width,*

$$\dot{M}_a = g_a w + \sum_b g_b u_{ab}, \quad \dot{u}_{ab} = g_a M_b + g_b M_a, \quad (51)$$

$$\dot{w} = 2 \sum_b g_b M_b, \quad G_{ab} = u_{ab} + \delta_{ab} w. \quad (52)$$

Theorem 16 (Finite-horizon deterministic wide limit). *For any finite realized recurrent matrix W , let all coordinates of b_s, b_w, c be mutually independent $\mathcal{N}(0, 1/N)$ and*

$$q_N = (M_s, M_w, u_{ss}, u_{sw}, u_{ww}, w), \quad q_0 = (0, 0, 1, 0, 1, 1). \quad (53)$$

For every $\epsilon > 0$,

$$\mathbb{P}(\|q_N(0) - q_0\|_\infty > \epsilon) \leq \min \left\{ 1, \frac{9}{N\epsilon^2} \right\}. \quad (54)$$

Fix a cue law with bounded second moments and define

$$g_a(M) = \mathbb{E}[z_a \sigma(-z \cdot M)], \quad A = \max_a \mathbb{E}|z_a|, \quad C = \frac{1}{4} \max_a \sum_b \mathbb{E}|z_a z_b|. \quad (55)$$

Then $\|g(M)\|_\infty \leq A$ and

$$\left\|g(M) - g(\widetilde{M})\right\|_\infty \leq C \left\|M - \widetilde{M}\right\|_\infty. \quad (56)$$

Let $q(t)$ be the deterministic solution from q_0 , let $c_\star = W(e^{-1}) = \sup_x x\sigma(-x)$, where W is Lambert's function, and set

$$R_H = 3 + 4c_\star H, \quad B_{H,r} = R_H + r, \quad L_{H,r} = 4(A + B_{H,r}C) \quad (57)$$

for a fixed deterministic radius $r > 0$. The reference Gram energy $R = w + u_{ss} + u_{ww}$ obeys $R(0) = 3$, $R' \leq 4c_\star$, and therefore $\|q(t)\|_\infty \leq R_H$ on $[0, H]$. The vector field is $L_{H,r}$ -Lipschitz in infinity norm on the radius- r tube about this path. If

$$\|q_N(0) - q_0\|_\infty \leq e^{-L_{H,r}H} \min\{r, \delta\}, \quad (58)$$

then q_N does not leave the tube before H and

$$\sup_{t \leq H} \|q_N(t) - q(t)\|_\infty \leq \delta. \quad (59)$$

Consequently,

$$\mathbb{P}\left(\sup_{t \leq H} \|q_N(t) - q(t)\|_\infty > \delta\right) \leq \min\left\{1, \frac{9e^{2L_{H,r}H}}{N \min\{r, \delta\}^2}\right\}. \quad (60)$$

For $r = 1$ and $0 < \delta \leq 1$, the denominator reduces to $N\delta^2$ with the deterministic constant $L_{H,1}$. Thus $q_N \rightarrow q$ uniformly on fixed horizons in probability, with initialization scale $O_{\mathbb{P}}(N^{-1/2})$. For the paired both/weak-only result, use a common r and the maximum of the two deterministic $L_{H,r}$ constants for the two fixed cue laws.

This theorem is a genuine large-width result at recurrence-invisible zero lag for any finite realized recurrent gain. It is not a positive-lag recurrent DMFT.

Corollary 17 (Zero-lag suppression and weak-only learnability). *For the noiseless positive rank-one cue law (zero cue noise), zero lag, zero background noise, and any finite realized recurrent matrix, use the deterministic symmetric initial state $(M_s, M_w, u_{ss}, u_{sw}, u_{ww}, w) = (0, 0, 1, 0, 1, 1)$. For every $\rho > 0$, the both-minus-weak gap and rate satisfy*

$$\bar{\Delta}(0) = 0, \quad \bar{d}(0) = 0, \quad \bar{d}'(0) = -\rho^2/2 < 0. \quad (61)$$

Hence $\bar{\Delta}(t) < 0$ for all sufficiently small $t > 0$: suppression begins at initialization and is not transfer-then-suppression. In the weak-only subsystem,

$$u_{ww} = w = \sqrt{1 + M_w^2}, \quad \dot{M}_w = 2\sigma(-M_w)\sqrt{1 + M_w^2} > 0, \quad (62)$$

so every finite $\beta > 0$ is reached and

$$\tau_\beta \leq \frac{\beta}{2\sigma(-\beta)}. \quad (63)$$

These identities and the target bound are independent of the arbitrary finite realized W . The finite-width outcome-suppression/learnability conclusion follows from Theorem 16 only after deterministic witness margins and its common tube event are established; ordinary floating-point integration is not silently promoted to such a proof.

7 Local mechanism-level control with Counterfactual Drift Correction

CDC uses the paired weak-only drift target to intervene on the diagnosed local mechanism; it is not presented as a general optimizer or robustness method. Flatten the trainable tensors into one fixed parameter chart with the standard product Euclidean/Frobenius inner product. Every gradient, dot product, norm, projection, Lipschitz ball, and straight segment in this section uses that chart and metric; no reparameterization-invariance or natural-gradient claim is made.

At a both-feature state θ , let

$$a = \nabla m_s(\theta), \quad p = \nabla m_w(\theta), \quad v = -\nabla L(\theta), \quad F_s = a \cdot v, \quad F_w = p \cdot v. \quad (64)$$

A same-initialization weak-only shadow supplies F_w^W . Define

$$q = \begin{cases} p - \frac{p \cdot a}{\|a\|^2} a, & a \neq 0, \\ p, & a = 0, \end{cases} \quad \delta = [F_w^W - F_w]_+, \quad \alpha = \begin{cases} \delta / \|q\|^2, & \delta > 0, q \neq 0, \\ 0, & \delta = 0. \end{cases} \quad (65)$$

The corrected velocity is $v' = v + \alpha q$. A positive deficit with $q = 0$ is infeasible. The theorems assume full-batch gradients, plain SGD, zero weight decay, and no clipping. A nonbinding coefficient cap leaves the uncapped result unchanged; a binding cap preserves orthogonality but generally loses target attainment, so no target-attaining optimality claim then applies. In the fixed parameter chart, feasible uncapped target-attaining CDC is the unique minimum- Euclidean-norm correction satisfying zero instantaneous strong-drift change and the weak-drift lower bound. Its finite-step guarantee is local and one-step.

Theorem 18 (CDC local guarantees). *The construction in Eq. (65) has the following fixed-coordinate properties.*

1. **Instantaneous preservation:** $a \cdot v' = a \cdot v$ exactly, including when α is capped.
2. **Minimum Euclidean norm:** for the feasible uncapped problem, $d^* = \alpha q$ is the unique minimum-norm, lower-bound-attaining correction satisfying $a \cdot d = 0$ and $p \cdot (v + d) \geq F_w^W$.
3. **Local finite-step bound:** if ∇m_s is L -Lipschitz in the same norm on a convex neighborhood containing both straight coordinate step segments, then

$$|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2). \quad (66)$$

For the active uncapped correction, if $\|v\| \leq V$, $\delta \leq D$, and $\|q\| \geq q_{\min} > 0$, the right side is at most

$$\frac{L}{2} \eta^2 [V^2 + (V + D/q_{\min})^2]. \quad (67)$$

This explicit norm bound also remains valid under a nonnegative upper cap, which can only reduce α .

For a positive raw deficit, exact feasibility can be stated intrinsically only as dm_w having a nonzero restriction to $\ker(dm_s)$; in this metric it is equivalent to $q \neq 0$. The represented q , its norm, and the selected minimum-norm correction are metric- and parameter-scale-dependent. The finite-step result is one-step and local in the same chart. It is not a trajectory guarantee and does not imply final strong-response retention.

8 Experimental protocol

8.1 Synthetic task and pairing

Sequences have two input channels. A strong cue of scale ρ and a unit weak cue appear at controlled positions separated by a lag. Positive tasks align both cues with the label; the negative control reverses the strong cue. Data and initialization are paired across both-feature and weak-only conditions. Training uses deterministic full-batch cross-entropy with zero weight decay. The nonlinear fixed-cell study trains the pair in lockstep so both direct response drifts are available at every logged time; deterministic conditions can otherwise be run sequentially without changing either trajectory.

8.2 Study designs

Table 5 summarizes model, task, seed, and compute settings. The controlled linear grid uses width 96, gain 0.5, four strengths, lags $\{0, 2, 4, 8\}$, and seed IDs 4–11. Each lag therefore has 32 strength–seed cells but only eight independent seed IDs. Its gate is $\beta = 0.25$ by $H = 5$; the negative control has eight seeds at $(\rho, \text{lag}) = (4, 4)$. The finite-width reference study uses widths 32–256, gain 0.5, and disjoint seeds for its empirical reference.

The nonlinear fixed-cell study uses width 32, $(\rho, \text{lag}) = (4, 2)$, eight seeds per architecture, gain 0.8, cue-noise standard deviation 0.1, zero background noise, and first-hit target $\beta = 0.5$. Tanh RNNs and GRUs use direct automatic-differentiation response drift. The broad tangent pilot uses the same cell on 16 architecture–seed records and evaluates five outputs. The held-out event test uses a fresh crossed design with four data seeds and eight model seeds per architecture and evaluates drift- and response-crossing labels.

The semi-real generated-cue study contains a 4×8 crossed design for each of MNIST 3-vs-8 and FashionMNIST 0-vs-6. It retains real core pixels and ablates only an exactly removable generated cue, so it is a semi-real intervention rather than a natural-spurious-correlation benchmark. The threshold-robust extension contains 192 nonlinear records across tanh and GRU, three fixed cells, and the first-hit grid $\beta \in \{0.25, 0.5, 1, 2\}$. The mechanism-control comparison contains 256 method-records across ERM, CDC, and adapted Bloop-style and PCGrad-style shadow-target variants. These variants use response-gradient shadow targets and are not claimed to reproduce the original algorithms. All shadow methods use a second weak-only model and therefore require roughly twice the model work.

8.3 Statistics and censoring

Independent seed pairs are the unit of replication only when neither seed axis is reused. The controlled linear grid reuses eight seed IDs across four strengths; the nonlinear 3/8, 5/8, and 0/8 proportions describe only the eight observed seeds per architecture. Neither design estimates universal event prevalence. For the held-out event test, two-way percentile intervals resample the observed data- and model-seed axes independently with 10,000 replicates. These intervals are conditional on the observed levels: an all-correct table necessarily yields $[1, 1]$ and does not quantify unseen failure modes.

Hitting times beyond the horizon are right-censored and never converted to finite delays. Conditional crossing-time means are reported with their crossing-record denominator. The exploratory nonlinear 5×5 grid uses seeds 0–3 and is reported only as a post-hoc search. The semi-real study uses dataset-specific two-way crossed-seed intervals; absent weak-only first hits censor causal interpretation in both directions. The threshold-robust extension uses a shared 5,000-replicate two-way bootstrap, equal-weights its three cells, and Holm-adjusts the two architecture tests; the resulting estimates are not used to rank architectures. The mechanism-control comparison uses the same bootstrap draws and equal-weights two cells. Its weak-rescue contrast is CDC minus comparator for $-|\text{weak AUC gap}|$ (positive favors CDC); trajectory and final contrasts are comparator minus CDC absolute deviation from paired ERM (positive favors CDC).

9 Results

9.1 Paired weak-AUC deficits and weak-only learnability

At lags 0, 2, and 4, all 32 ρ -seed cells per lag (four strengths by eight seed IDs) have positive paired weak-AUC deficits and learnable weak-only controls under the specified gate. Averaged across ρ , the lag-wise deficits are 5.76, 8.93, and 6.11, respectively; the measured effect peaks at lag 2 rather than decreasing monotonically with lag. At every fixed lag, the mean deficit increases monotonically over the measured ρ grid. The negative-control cell is transfer in 8/8 seeds, with mean gap -4.038 and 95% CI $[-4.498, -3.578]$. Every lag-8 cell fails the weak-only gate: all 32 are weak-only nonlearnable and causally indeterminate under $\beta = 0.25, H = 5$. They supply neither causal-starvation certificates nor evidence that starvation is absent. Final accuracy is 1.0 throughout, and no row meets the target at initialization.

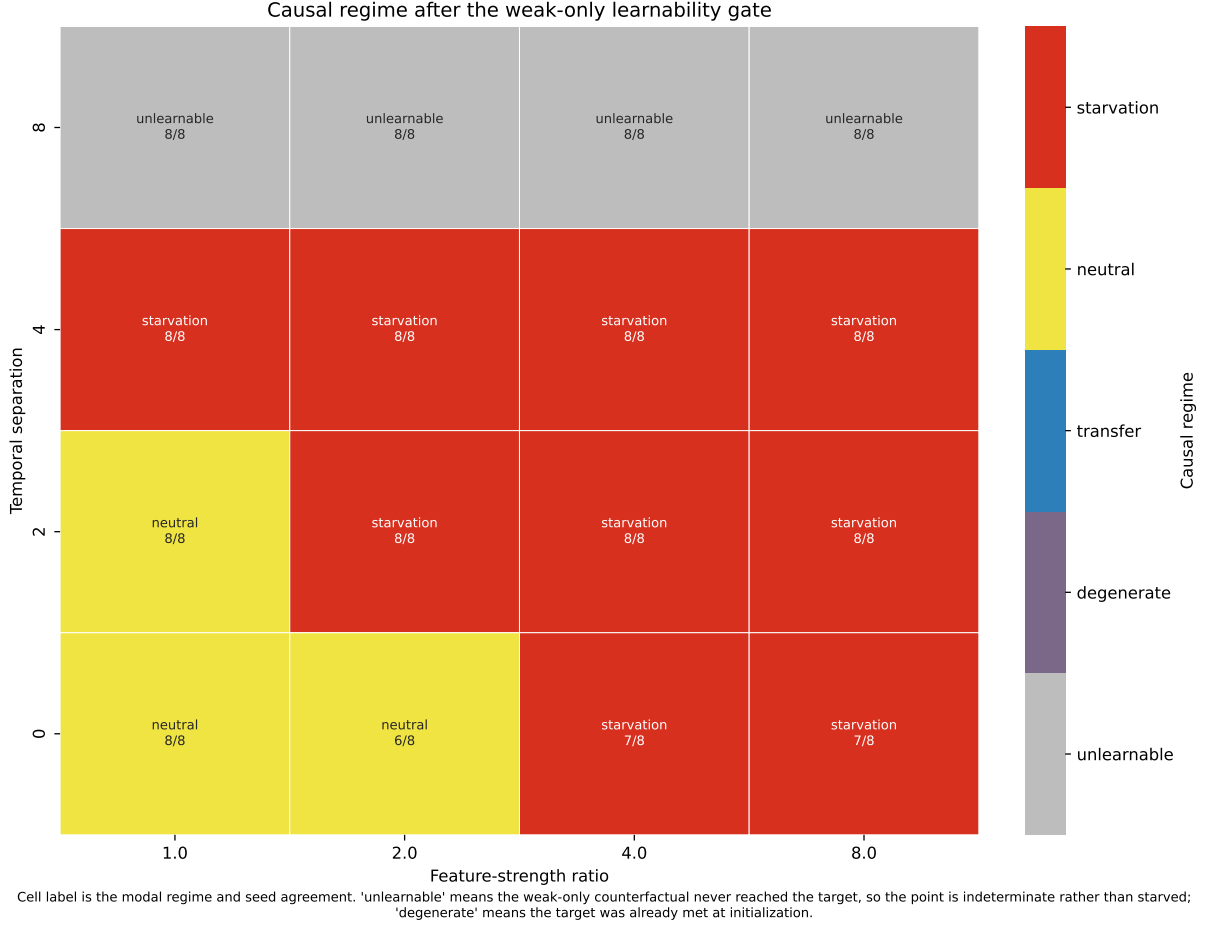


Figure 1: Controlled dense-linear paired weak-AUC deficits and weak-only gate status. The specified learnability gate makes all lag-8 points causally indeterminate rather than positive or negative starvation evidence.

9.2 Empirical width trends and exact zero-lag checks

The finite-width reference study compares finite networks with a disjoint-seed empirical *closure reference*; it is not an independently derived theory. At widths 32, 64, 128, and 256, overall both-condition trajectory NRMSE is 0.09084, 0.04761, 0.05149, and 0.02958, respectively; weak-only NRMSE is 1.56931, 0.38792, 0.40201, and 0.20024. Both improve end-to-end but are non-monotone at $N = 128$. All 32 point-condition-diagnostic combinations improve end-to-end; large normalized weak-only values at difficult points have near-zero denominators. The exact finite-width geometry check has zero recorded relative error and the maximum projected-identity error is 2.86×10^{-6} .

In the exact zero-lag discretization check, the time-discretization residual scales as $O(\eta)$ with fitted slope 1.0115. A frozen-geometry integrator agrees with the projected-flow integrator to 1.91×10^{-15} . These checks concern the proved zero-lag special case and do not validate a general recurrent mean-field closure.

9.3 Tanh certificates versus ungated GRU crossings

Table 1 is the authoritative nonlinear result at the fixed $(\rho, \text{lag}) = (4, 2)$ cell. Tanh has a complete exact-drift/response/tail- area transfer-to-starvation certificate in $3/8$ observed seeds; the other five have negative exact relative drift from initialization and therefore do not exhibit the transfer-to-starvation path type. GRU has exact drift and response crossings in $5/8$ seeds, but every weak-only trajectory fails to reach $\beta = 0.5$; its

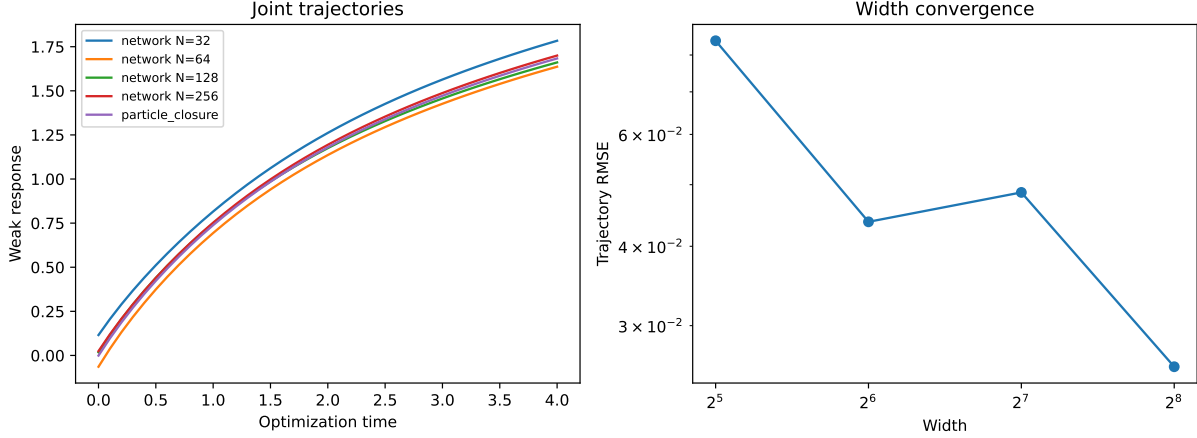


Figure 2: Finite-width agreement with a held-out empirical finite-network closure reference. This is not DMFT validation.

Table 1: Final nonlinear fixed-cell study result at $\rho = 4$, lag 2, eight seeds. Crossing times and confidence intervals are conditional on crossing and must not be read as eight-seed estimates.

Model	exact drift	response	causal	mean τ_d	mean τ_r	max $ R_w^B - R_w^W $
tanh	3/8	3/8	3/8	0.194	0.389	0.02197
GRU	5/8	5/8	0/8	3.846	7.200	0.002986

causal count is therefore 0/8 and causal status is indeterminate under that gate. This separates rate/outcome suppression from causal starvation; it does not rank architectures or estimate prevalence. The nonzero direct-versus-projected residuals empirically confirm that Gg cannot be silently treated as the exact nonlinear probe derivative.

The prespecified gate is the first hit at $\beta = 0.5$, but a single binary operating point can hide whether failures are marginal. We therefore add a post-hoc terminal analysis using $B_{W,s}(H) = M_{w,s}^W(H)$ and the empirical fraction of seeds with $B_{W,s}(H) \geq \beta$ over candidate thresholds. No model was retrained: Figure 3 is derived from the recorded per-seed nonlinear summary. Tanh terminal responses range from 4.413 to 4.676 (mean 4.497), with 8/8 above $\beta = 0.5$; GRU responses range from 0.0828 to 0.187 (mean 0.130), with 0/8 above 0.5. Thus the GRU weak-only gate failure is not marginal at that operating point, but it still provides neither a causal-starvation certificate nor evidence that starvation is absent. Terminal reach is not generally interchangeable with the prespecified first-hit event on a nonmonotone trajectory, and this post-hoc diagnostic supports neither a new endpoint nor an architecture ranking.

The post-hoc tanh search over $\rho \in \{1.5, 2, 3, 4, 6\}$ and lags $\{0, 1, 2, 4, 8\}$ found no cell with 4/4 causal certificates; the best cells reached 3/4. This negative exploratory result is not used to select the reported fixed cell.

9.4 Broad tangent pilot and held-out event test

The initialization-frozen tangent surrogate was first evaluated on one data seed and eight model seeds per architecture. It correctly classified phase, drift crossing, response crossing, causal certificate, and weak-only learnability in 12/16, 14/16, 14/16, 14/16, and 12/16 records. The five-output criterion nevertheless failed because tanh weak-only learnability was 4/8, below the prespecified 5/8 floor.

A fresh factorial then crossed four data seeds with eight model seeds for each architecture and restricted evaluation to crossing events. Table 2 reports classification correctness rather than observed crossing prevalence. Response-crossing labels were correct in 64/64 records and drift-crossing labels in 60/64; their two-way-bootstrap intervals were $[1, 1]$ and $[0.8125, 1]$.

The restriction is substantive. In the factorial, phase, causal-certificate, and learnability correctness were

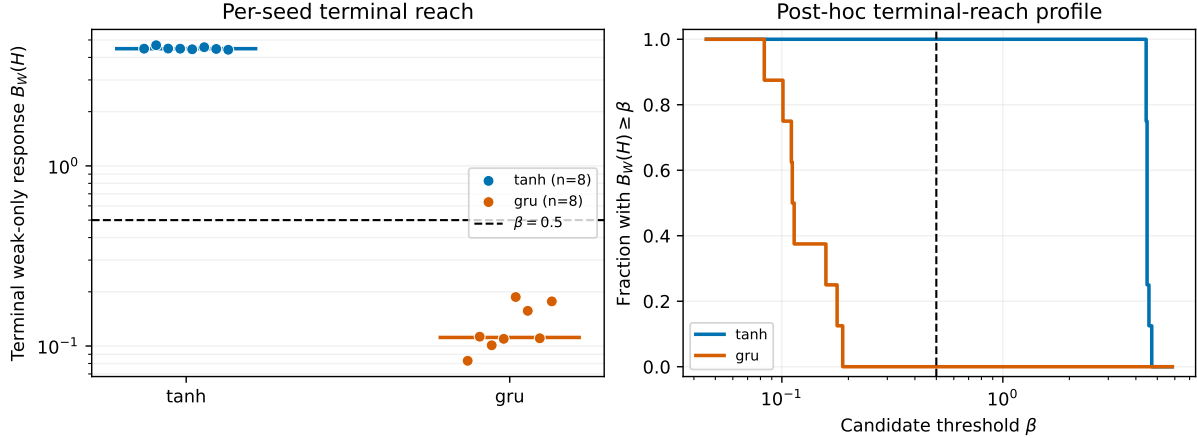


Figure 3: Post-hoc weak-only terminal-response analysis. Points are terminal responses for all eight seeds; horizontal segments mark medians and the dashed line marks the prespecified $\beta = 0.5$. The right panel sweeps candidate thresholds without rerunning training. Terminal reach is not generally interchangeable with a first-hit gate for nonmonotone trajectories.

Table 2: Tangent-surrogate evaluations. The broad five-output pilot failed; the fresh crossed-seed test supports only drift- and response-crossing event classification. Intervals resample data- and model-seed axes.

Protocol	Endpoint	Overall	tanh	GRU	95% CI
Broad pilot	phase	12/16	6/8	6/8	diagnostic
Broad pilot	weak learnability	12/16	4/8	8/8	failed
Held-out 4×8	drift event	60/64	32/32	28/32	$[0.8125, 1]$
Held-out 4×8	response event	64/64	32/32	32/32	$[1, 1]$

only 52/64, 56/64, and 49/64, including tanh learnability of 17/32. Crossing times were systematically early (overall drift/response bias $-0.822/-1.189$), while tanh response-gap, weak-response, and drift RMSE were 2.455, 2.489, and 0.490. The result is therefore an uncalibrated event classifier at one width/task cell. It neither proves Theorem 5’s kernel premise nor validates phase, causal status, learnability, trajectory magnitude, calibrated time, or prevalence.

9.5 Dense mechanism-control comparison

All five shadow-based methods reduce the weak AUC gap from ERM’s 10.5155 to approximately zero and attain their shared weak-drift target at every logged step. Their weak outcomes are numerically similar to roughly four decimal places, though 9/10 direct paired comparisons are significant at 0.05; these differences are small relative to the approximately 10.5 change from ERM. CDC alone preserves instantaneous strong drift to 4.77×10^{-7} , versus 1.48×10^{-1} to 3.75×10^{-1} for the alternative shadow constraints. This local property does not yield the best final strong response: unconstrained rescue and the adapted PCGrad-style shadow-target method end at 1.295, compared with CDC’s 1.042 and ERM’s 1.364.

9.6 Semi-real, threshold-robust, and mechanism-control extensions

The semi-real generated-cue study contains 64 records. MNIST 3-vs-8 and FashionMNIST 0-vs-6 each have 0/32 weak-only-learnable records and therefore 0/32 causal certificates under the specified gate. The mean weak-AUC gap is $-1.2362842136667493 \times 10^{-3}$ (95% CI $[-3.4405428466004646, 1.0159591371180453] \times 10^{-3}$) for MNIST and $-1.0921728159018996 \times 10^{-4}$ (95% CI $[-4.129384520886106, 3.182670049955049] \times 10^{-3}$) for FashionMNIST. Positive values would mean that weak-only core gain exceeds both-feature core gain.

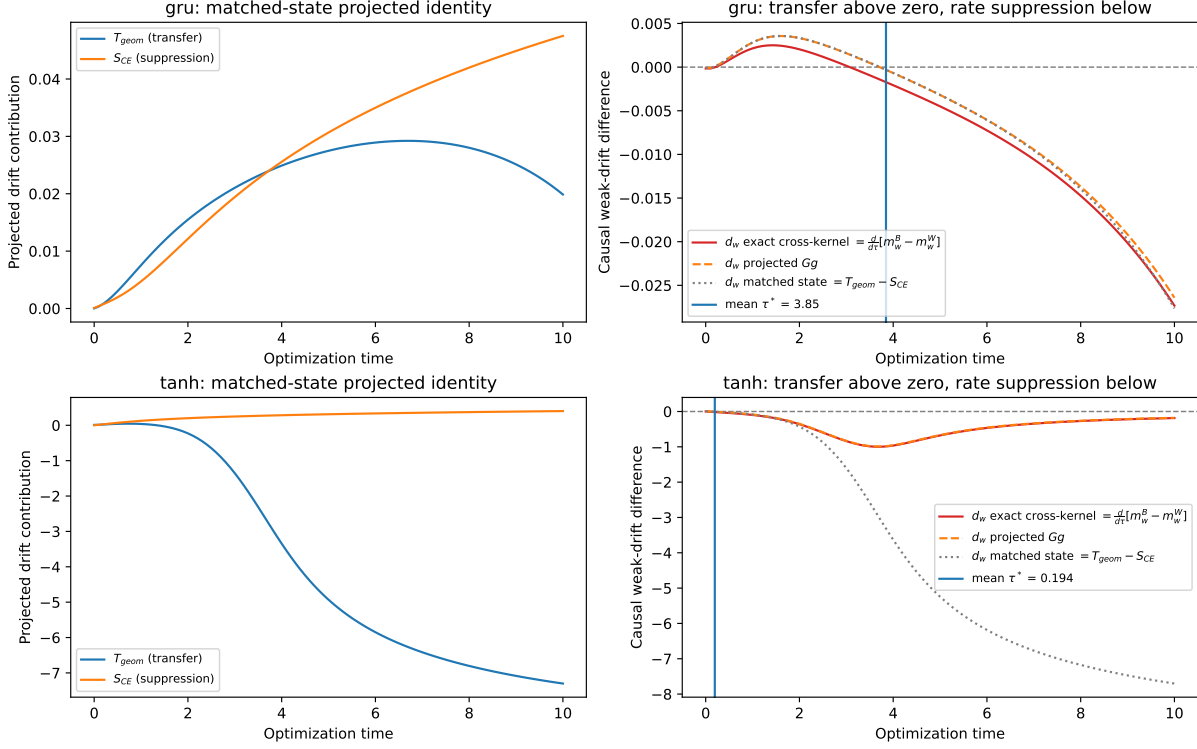


Figure 4: Paired nonlinear diagnostic trajectories. Direct-autograd drift defines the reported events; projected and matched-state quantities are separately styled diagnostics.

Because every weak-only first-hit gate fails, the zero causal counts are negative/indeterminate: they are not evidence that starvation is absent and do not show that either dataset is intrinsically unlearnable.

The threshold-robust extension contains 192 nonlinear records. Its equal-weight three-cell all- β causal-certificate macro estimate is 0.292 for tanh (95% CI [0.167, 0.417]) and 0 for GRU (95% CI [0, 0]). The exact values appear in Table 7. The Holm-adjusted one-sided p -value is 1.0 for each. The threshold-robust endpoint is negative and supports no architecture ranking.

The mechanism-control comparison contains 256 method-records. Weak rescue is $-|\text{weak AUC gap}|$, so CDC minus comparator is higher-better. Trajectory and final deviations are absolute deviations from paired ERM, so comparator minus CDC is positive when CDC preserves the strong response better. Table 4 reports the equal-weight two-cell contrasts. The joint weak-rescue/strong-preservation criterion is true only against the adapted Bloop-style shadow-target variant and false against the adapted PCGrad-style shadow-target variant. CDC therefore does not beat the latter under the joint criterion. Because both are adapted response-gradient shadow-target variants, these results establish neither parity with the original methods nor broad CDC superiority.

10 Discussion and limitations

Causal diagnosis and empirical scope. The empirical findings are deliberately regime-specific. The controlled linear grid shows positive paired weak-AUC deficits with learnable weak-only controls in all 32 strength-seed cells per lag at lags 0, 2, and 4, but those cells reuse eight seed IDs. All 32 lag-8 cells fail the gate under $\beta = 0.25, H = 5$ and are causally indeterminate. At the nonlinear fixed cell, tanh has 3/8 complete certificates; GRU has 5/8 drift and response crossings but 0/8 causal certificates because every weak-only path fails $\beta = 0.5$. The terminal-threshold profile shows that this GRU failure is not marginal at that operating point, but it neither replaces the first-hit gate nor ranks architectures. The semi-real and threshold-robust extensions are negative or indeterminate. None of these fixed-grid counts estimates universal

Table 3: Dense CDC ablation, eight paired seeds. Shadow-based arms have access to a matched weak-only model; the other arms do not. “Strong change” is the maximum absolute instantaneous strong-drift perturbation.

Method	Shadow	Weak AUC gap	Strong change	Final m_s
ERM	no	10.5155	—	1.364
Spectral Decoupling	no	9.0229	—	0.829
Interaction	no	10.5145	—	1.365
CDC	yes	−0.0005	4.77×10^{-7}	1.042
Loss-gradient projection	yes	−0.0002	3.75×10^{-1}	0.371
Unconstrained rescue	yes	−0.0007	1.48×10^{-1}	1.295
Adapted Bloop-style shadow-target variant	yes	0.0002	3.75×10^{-1}	0.370
Adapted PCGrad-style shadow-target variant	yes	−0.0007	1.48×10^{-1}	1.295

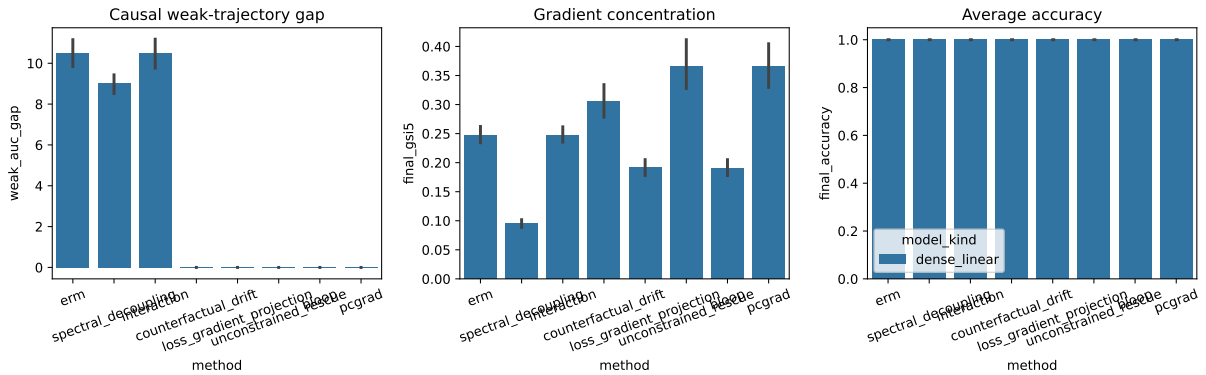


Figure 5: Eight-method dense-linear mitigation ablation. CDC’s established distinction is instantaneous first-order preservation, not superior weak rescue or final strong retention.

prevalence or an architecture ordering.

Exact dynamics and pathwise attribution. The finite-width identity $\dot{M}_w = c_w^\top q(r)/n$ separates cross-entropy sample weighting $q(r)$ from response-specific geometry c_w without a mean-field assumption. Together with exact sample-logit dynamics and the matched intervention, it identifies the sources of weak-response motion without silently imposing a two-response closure. The balance $\Delta = P - N$ then shows that outcome suppression occurs only when accumulated suppression overtakes accumulated help. Neither rate nor outcome suppression is by itself causal starvation; weak-only learnability remains an independent requirement.

Prediction limits. The realized-initialization $d(0)$ criterion gives an exact local sign only in its stated dense-linear setting. The conditional parameter-tube and rank-one criteria, transverse hitting-time bound, and recurrence-invisible zero-lag wide limit require their explicit assumptions. The zero-lag result permits any finite realized recurrent matrix but requires zero lag and zero background input. Stronger positive-lag parameter-level prediction from ρ , lag, and recurrent geometry remains unresolved because no nonvacuous general product-ball or kernel-movement constants are constructed.

The initialization-frozen tangent surrogate retains nonlinear cross-entropy but is exact only for its own tangent model. Its use for trained tanh RNNs or GRUs assumes instantaneous or secant-kernel movement control that is not proved here. The broad pilot fails its five-output criterion, while the held-out test supports only crossing-event classification at one cell. Its timing predictions are early and its tanh trajectory magnitudes are poor. Event accuracy therefore does not prove kernel stability, calibrated dynamics, or causal prediction.

Table 4: Equal-weight two-cell mechanism-control contrasts over 256 method-records. Both comparators are adapted shadow-target variants. Brackets are 95% bootstrap intervals.

Comparator	CDC-comparator weak rescue	Comparator-CDC trajectory dev.	Comparator-CDC final dev.	Joint
Adapted Bloop-style variant	0.002198250399210656 [0.0009779187294930126, 0.003277366267916477]	3.4651361294978416 [3.2443827215131754, 3.5849715262780952]	0.6422362388111651 [0.6288088704226539, 0.6563168084365315]	true
Adapted PCGrad-style variant	-0.00015351238407674823 [-0.00047909348592838785, 0.00013248012419808214]	-0.48709889128076556 [-0.6897378944428417, -0.2737932373499074]	-0.2565436437726021 [-0.3016841153614223, -0.208765183574514]	false

Intervention and external-validity limits. In a fixed Euclidean/Frobenius parameter chart, feasible uncapped target-attaining CDC is the unique minimum-norm correction that preserves instantaneous strong drift and meets the weak-drift lower bound. The finite-step result is local and one-step. The dense comparison exhibits this instantaneous distinction, but CDC does not have the best final strong response. The broader joint criterion passes only against the adapted Bloop-style variant and fails against the adapted PCGrad-style variant. All shadow methods require a constructible ablation, a second model, and full-batch gradients; no parity with the original comparator algorithms or broad CDC superiority follows.

The experiments cover controlled synthetic families and one semi-real generated- cue intervention over fixed cells and seed grids. The image study is not an unmodified natural-spurious-correlation benchmark. No Waterbirds or natural- spurious-correlation claim is made. General recurrent causal prediction, calibrated nonlinear crossing times, architecture rankings, and broad intervention performance remain open.

11 Conclusion

Weak learning failure alone is not gradient starvation. A small weak response can reflect weak-only gate failure, altered cross-entropy weighting, response-specific geometry, suppression too brief to reverse earlier help, or causally gated outcome suppression. The matched same-initialization intervention separates these cases. Exact finite-width dynamics give $\dot{M}_w = c_w^\top q(r)/n$, and the pathwise identity $\Delta = P - N$ states exactly when accumulated suppression exceeds earlier transfer. Causal starvation additionally requires an independently learnable weak-only counterfactual; gate failure establishes neither starvation nor its absence.

Prediction remains local or conditional. Realized $d(0)$ and proved local jets give assumption-scoped signs, while global positive-lag parameter-level prediction remains unresolved. The frozen tangent surrogate supplies an empirical crossing- event heuristic absent kernel-movement bounds, not calibrated or causal dynamics. CDC supplies a unique fixed-chart local correction, not a trajectory-level or broadly superior optimizer.

The experiments support the diagnostic separation rather than a universal law: controlled linear regimes can yield consistent causal deficits, while long-lag, nonlinear, and semi-real regimes may fail the weak-only gate or show only uncertified crossings. The strongest supported chain is exact dynamics $\rightarrow$ matched causal intervention $\rightarrow$ pathwise attribution $\rightarrow$ local mechanism-level control.

A Proofs for exact dynamics and causal-separation results

A.1 Proof of Proposition 1 and Corollary 2

For $\ell(r) = \log(1 + e^{-r})$, $\ell'(r) = -\sigma(-r)$. Therefore

$$\dot{\theta}_C = -\nabla L_C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \nabla r_i^C. \quad (68)$$

The chain rule gives

$$\dot{M}_a^C = \langle \nabla M_a^C, \dot{\theta}_C \rangle = \frac{1}{n_C} \sum_i \langle \nabla M_a^C, \nabla r_i^C \rangle \sigma(-r_i^C), \quad (69)$$

which proves Proposition 1. Under exact realization, $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C$. Substitution and exchange of finite sums yield Eq. (8). If an orthogonal residual logit term is present, substitution of $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C + \nabla r_i^\perp$ instead yields Eq. (9). $\square$

A.2 Proof of Corollary 3 and Theorem 5

The parameter flow above can be written $\dot{\theta} = J^\top q(r)/n$. Therefore the chain rule gives $\dot{r} = J\dot{\theta} = JJ^\top q(r)/n$ and $\dot{M} = \langle \nabla M, \dot{\theta} \rangle = c^\top q(r)/n$, proving Corollary 3.

For the comparison, q is $1/4$ -Lipschitz componentwise and hence in Euclidean norm. Subtracting Eq. (12) from Eq. (11), adding and subtracting $\Theta_0 q(r)$, and writing $e_r = \|r - \bar{r}\|$ gives

$$e'_r(u) \leq \frac{\|\Theta_0\|}{4n} e_r(u) + \frac{B_\Theta V}{n} = a e_r(u) + \frac{B_\Theta V}{n}. \quad (70)$$

Grönwall with $e_r(0) = 0$ proves Eq. (14). The same add-subtract step gives

$$|\dot{M} - \dot{M}| \leq \frac{B_c V}{n} + \frac{\|c_0\|}{4n} e_r. \quad (71)$$

Integrating and using $\int_0^t \phi_a(u) du = \psi_a(t)$ proves Eq. (15).

For SGD, the fundamental theorem of calculus along the straight update segment gives

$$r_{j+1} - r_j = \tilde{J}_j \delta \theta_j = \frac{\eta}{n} \tilde{\Theta}_j q(r_j), \quad (72)$$

$$M_{j+1} - M_j = \langle \tilde{m}_j, \delta \theta_j \rangle = \frac{\eta}{n} \tilde{c}_j^\top q(r_j). \quad (73)$$

Subtracting the frozen Euler recursion yields scalar recurrences

$$e_{j+1} \leq (1 + \eta a) e_j + \frac{\eta D_\Theta V}{n}, \quad (74)$$

$$E_{j+1} \leq E_j + \frac{\eta D_c V}{n} + \frac{\eta \|c_0\|}{4n} e_j. \quad (75)$$

Solving the first geometric recurrence and summing it in the second gives Eq. (19), including the stated $a = 0$ limits. Paired bounds are the triangle inequality. Error-separated sign transfer follows immediately; first-hit transfer requires the additional assumptions of Corollary 14. $\square$

A.3 Proof of the dense-linear geometry

For a unit response mode in channel a at lag ℓ_a , write b_a for the corresponding input column. Direct differentiation of $c^\top W^{\ell_a} b_a$ gives gradient $p_a = W^{\ell_a} b_a$ with respect to c , gradient $q_a = (W^{\ell_a})^\top c$ with respect to column b_a , and recurrent gradient

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (76)$$

The parameter blocks are disjoint, so their gradient inner products add. Input columns contribute only when the channels coincide, giving Eq. (21). $\square$

A.4 Proof of Theorem 6

For the positive rank-one cue laws, $g^B = s_B(\rho, 1)$ and $g^W = s_W(0, 1)$ at the shared initialization. Corollary 2 and the exact dense geometry give

$$\dot{M}_w^B(0) = s_B(\rho G_{ws} + G_{ww}), \quad \dot{M}_w^W(0) = s_W G_{ww}. \quad (77)$$

Subtraction, with $K = G_{ws}$ and $A = G_{ww}$, proves Eq. (27). Since $\rho s_B > 0$, rearrangement gives Eq. (28). The displayed K, A, S_w formulas are Eq. (21) specialized to a zero-lag strong mode and an ℓ -lag weak mode. Under the iid width scaling at fixed positive ℓ , expanding the gates at zero gives

$$d_0 = \rho \left(\frac{K}{2} - \frac{A M_s(0)}{4} \right) + O_{\mathbb{P}}(N^{-1}), \quad (78)$$

whose leading sign is centered and realization-dependent. Shared initialization also gives $\Delta(0) = 0$ and $\Delta'(0) = d_0$. Therefore $\Delta(t)/t \rightarrow d_0$ as $t \downarrow 0$: a strict negative (positive) d_0 forces strict local outcome suppression (transfer) on some sufficiently small positive interval. This limit establishes no quantitative endpoint, and when $d_0 = 0$ first order is silent. This proves the exact criterion, its strongest unconditional local outcome implication, and the stated obstruction to a distribution-only positive-lag boundary. $\square$

A.5 Proof of Proposition 7 and Corollaries 8–9

The positive and negative parts satisfy $d = d_+ - d_-$ almost everywhere. Absolute continuity and the fundamental theorem of calculus therefore give

$$\Delta(t) = \Delta(0) + \int_0^t d(u) du = P(t) - N(t), \quad (79)$$

proving Proposition 7 and its pointwise sign equivalences. Under Corollary 8's sign pattern, $d_- = 0$ before τ_d and $d_+ = 0$ afterward. Hence $P(t) = P(\tau_d) = A_+$ and

$$\Delta(t) = A_+ - N(t), \quad t \geq \tau_d. \quad (80)$$

The post-crossing sign assumption makes N continuous and strictly increasing on every nontrivial post-crossing interval. Later equality, strict negativity, and uniqueness are therefore respectively equivalent to $N(H) \geq A_+$, $N(H) > A_+$, and strict post-crossing growth of N .

If $d \leq -\kappa$ after t_0 , then $N(t) - N(t_0) \geq \kappa(t - t_0)$; solve for the time needed to cover the remaining area. If $d'(t) \leq -\lambda$ and $d(\tau_d) = 0$, integration gives $d(t) \leq -\lambda(t - \tau_d)$ and then $N(t) - N(\tau_d) \geq \lambda(t - \tau_d)^2/2$. Since $N(\tau_d) = 0$ under the single-crossing hypotheses, applying Corollary 8 proves both stated bounds. $\square$

A.6 Proof of Theorem 10 and Corollary 11

Before exit from U_R , path length is at most $S_R t$, so exit cannot occur before $h_R = R/S_R$. Along this interval, $d' = j$, and integration of Eq. (34) gives

$$d_0 - \Lambda t \leq d(t) \leq d_0 - \lambda t. \quad (81)$$

The lower bound is positive before d_0/Λ , the upper bound is negative after d_0/λ , continuity gives a zero between them, and $d' < 0$ makes it unique. This proves Eq. (35). Integrating the upper bound from $\Delta(0) = 0$ gives

$$\Delta(t) \leq d_0 t - \frac{\lambda t^2}{2} < 0 \quad \text{for } t > 2d_0/\lambda. \quad (82)$$

For the required nondegenerate case $\beta > M_w^W(0)$, integrating the weak-only lower drift bound reaches β by the positive time in Eq. (36), proving the causal conclusion. If $\beta \leq M_w^W(0)$, attainment occurred before optimization and supplies no learnability gate.

For the safe case, $|j| \leq J$ gives $d(t) \geq d_0 - Jt$; one more integration gives Eq. (37). The strict premise $d_0 - Jh_R > 0$ makes both quantities positive on the certified horizon. $\square$

A.7 Proof of Theorem 12 and Corollary 13

The deterministic cue law gives $g^B = s_B(\rho, 1)$ and $g^W = s_W(0, 1)$. Equation (8) therefore gives $F_w^B = s_B H_B$, $F_w^W = s_W H_W$, $u'_B = s_B Q_B$, and $u'_W = s_W H_W$. Since

$$\frac{d}{du} \log \sigma(-u) = -(1 - \sigma(-u)), \quad (83)$$

differentiating Ψ yields Eq. (43). When both drifts are positive, monotonicity of the logarithm gives $\text{sign}(F_w^B - F_w^W) = \text{sign}(\Psi)$.

Under $\Psi' \leq -\kappa$, integration gives $\Psi(t) \leq \Psi(t_0) - \kappa(t - t_0)$. Continuity forces a zero by $t_0 + \Psi(t_0)/\kappa \leq H$, while strict decrease gives uniqueness. The primitive inequalities imply the derivative bound term by term. $\square$

A.8 Proof of Corollary 14

Continuity and first-entry semantics imply $x(\tau) = \beta$. Absolute continuity and $x' \geq \kappa$ almost everywhere give, by integration,

$$x(t) \leq \beta - \kappa(\tau - t) \quad (\tau - r \leq t \leq \tau), \quad x(t) \geq \beta + \kappa(t - \tau) \quad (\tau \leq t \leq \tau + r). \quad (84)$$

On $[0, \tau - r]$, the quantitative prehistory margin and $\epsilon < \eta$ give $x_N < \beta$. On $[\tau - r, \tau - \epsilon/\kappa]$, the left cone and the uniform error again give $x_N < \beta$. Hence $\tau_N \geq \tau - \epsilon/\kappa$. At $t = \tau + \epsilon/\kappa$, the right cone gives $x_N(t) \geq \beta$, so the hit is finite and $\tau_N \leq \tau + \epsilon/\kappa$. This proves Eq. (48).

For sampled data, the first grid point at or after $\tau + \epsilon/\kappa$ is at most h later and, by the radius condition, remains inside the right cone; it is therefore at or above β . No sampled hit can precede the continuous lower bound. If the piecewise-linear interpolant first crosses on $[t_{j-1}, t_j]$, then $x_N(t_j) \geq \beta$, so $t_j \geq \tau - \epsilon/\kappa$ and $t_{j-1} \geq \tau - \epsilon/\kappa - h$; its upper bound is no later than the sampled hit. This proves Eq. (49). Applying the bound separately to two finite hits and using the triangle inequality proves the paired statement. $\square$

A.9 Proof of Theorems 15 and 16

At zero lag and zero background input, every cue occurs at the final step. From $h_0 = 0$, induction gives zero hidden state before that step, so for every finite realized W ,

$$h_T = Wh_{T-1} + Bx_{T-1} = Bx_{T-1}. \quad (85)$$

The loss is independent of W , each recurrent-gradient contribution contains a zero incoming state, and therefore $\nabla_W L = 0$ throughout the unregularized flow. No distributional or zero-gain premise on W is needed. Gradient flow for the remaining parameters gives $\dot{b}_a = g_a c$ and $\dot{c} = \sum_b g_b b_b$. Differentiating $M_a = c \cdot b_a$, $u_{ab} = b_a \cdot b_b$, and $w = c \cdot c$ proves Eq. (52). The mode gradient has one component in the c block and one in its own b_a block, yielding $G_{ab} = u_{ab} + \delta_{ab} w$.

At initialization, the variances of M_s, M_w, u_{sw} are $1/N$ and those of $u_{ss} - 1, u_{ww} - 1, w - 1$ are $2/N$. Coordinatewise Chebyshev bounds and a union bound sum to $1 + 1 + 1 + 2 + 2 + 2 = 9$, proving Eq. (54).

The field bound $\|g(M)\|_\infty \leq A$ follows from $0 \leq \sigma \leq 1$. Since σ is $1/4$ -Lipschitz,

$$|g_a(M) - g_a(\widetilde{M})| \leq \frac{1}{4} \sum_b \mathbb{E}|z_a z_b| \left\| M - \widetilde{M} \right\|_\infty, \quad (86)$$

which proves Eq. (56). Define the Gram energy $R = w + u_{ss} + u_{ww}$. Along the deterministic flow,

$$R' = 4g \cdot M = 4\mathbb{E}[X\sigma(-X)], \quad X = z \cdot M. \quad (87)$$

Because $x\sigma(-x) \leq c_* = W(e^{-1})$ and $R(0) = 3$, $R(t) \leq R_H$. Every coordinate of q is a squared norm or inner product of b_s, b_w, c , so Cauchy-Schwarz gives $\|q(t)\|_\infty \leq R(t) \leq R_H$.

Consider two states in the cube $\|q\|_\infty, \|\tilde{q}\|_\infty \leq B_{H,r}$ and set $d = \|q - \tilde{q}\|_\infty$. Every product in the M and u equations satisfies the direct estimate

$$|g_a(M)y - g_a(\widetilde{M})\tilde{y}| \leq (A + B_{H,r}C)d. \quad (88)$$

There are at most three such products in an M equation and two in a u equation; the factor two and two cue indices give four in the w equation. Thus the full vector field is $L_{H,r} = 4(A + B_{H,r}C)$ -Lipschitz on the cube and therefore on the radius- r reference tube.

Let T_r be the first time before H that $\|q_N(t) - q(t)\|_\infty$ reaches the tube boundary. Up to T_r , Gronwall's inequality and Eq. (58) give

$$\|q_N(t) - q(t)\|_\infty \leq e^{L_{H,r}t} \|q_N(0) - q_0\|_\infty. \quad (89)$$

If $L_{H,r} > 0$, the initialization event makes this bound strictly less than r for every $t < H$, so the boundary cannot be reached before H . If $L_{H,r} = 0$, then $A = C = 0$, the cue law is zero almost surely, and the six-scalar vector field vanishes; the initial discrepancy is constant and cannot exit the closed tube. The endpoint may lie on the boundary. In either case the same estimate is at most δ throughout $[0, H]$, proving Eq. (59). Applying Eq. (54) with $\epsilon = e^{-L_{H,r}H} \min\{r, \delta\}$ proves Eq. (60). For the shared-initialization pair, use a common r and the maximum deterministic constant from the two fixed cue laws; the same concentrating initial six scalars control both paths. $\square$

A.10 Proof of Corollary 17

Under the stated noiseless positive rank-one cue law, the recurrence-invisible six-scalar equations give equal initial weak drifts in the two conditions for every finite realized W . Differentiating those drifts once at the symmetric state gives

$$(\dot{M}_w^B)'(0) = -\frac{\rho^2 + 1}{2}, \quad (\dot{M}_w^W)'(0) = -\frac{1}{2}, \quad (90)$$

so $\bar{d}'(0) = -\rho^2/2$. Taylor's theorem and continuity then make $\bar{\Delta}(t) < 0$ for all sufficiently small positive t .

The weak-only symmetric subsystem preserves $u_{ww} = w$. Direct differentiation shows $u_{ww}^2 - M_w^2 = 1$, yielding $u_{ww} = w = \sqrt{1 + M_w^2}$ and the displayed positive drift. While $0 \leq M_w \leq \beta$, it obeys $M_w \geq 2\sigma(-\beta)$; integration proves Eq. (63). The finite-width statement is then exactly Theorem 16 applied at deterministic suppression and target-attainment witness times with margins smaller than both strict witness margins. $\square$

B Proofs for local Counterfactual Drift Correction

B.1 Instantaneous preservation

If $a = 0$, both strong drifts vanish. Otherwise,

$$a \cdot q = a \cdot p - \frac{p \cdot a}{\|a\|^2}(a \cdot a) = a \cdot p - p \cdot a = 0, \quad (91)$$

so $a \cdot v' = a \cdot v + \alpha a \cdot q = a \cdot v$. This argument is independent of α , hence survives a cap. $\square$

B.2 Unique minimum-norm correction

Let $\bar{\delta} = F_w^W - F_w$ be the raw deficit and $\delta = [\bar{\delta}]_+$. If $\bar{\delta} \leq 0$, $d = 0$ is feasible and uniquely norm-minimal. Suppose $\bar{\delta} > 0$, so $\delta = \bar{\delta}$. Any feasible d obeys $a \cdot d = 0$ and $p \cdot d \geq \delta$. Because $p = q + (p \cdot a / \|a\|^2)a$ when $a \neq 0$ (and $q = p$ when $a = 0$), $p \cdot d = q \cdot d$. Cauchy-Schwarz in the fixed Euclidean subspace orthogonal to a gives

$$\delta \leq q \cdot d \leq \|q\| \|d\|, \quad \|d\| \geq \delta / \|q\|. \quad (92)$$

Equality requires a nonnegative multiple of q . The correction $d^* = (\delta / \|q\|^2)q$ is feasible and attains this lower bound, so it is the unique minimum-norm, lower-bound-attaining feasible correction. Larger positive multiples can also attain the target but have larger norm. A positive deficit is feasible exactly when $q \neq 0$ in this representation. A nonbinding cap leaves d^* unchanged; a binding cap generally loses target attainment, so the target-attaining optimization claim no longer applies. $\square$

B.3 Local finite-step deviation

In the fixed chart and norm, L -Lipschitzness of the gradient on both straight segments gives the smoothness remainder

$$|m_s(\theta + h) - m_s(\theta) - \nabla m_s(\theta) \cdot h| \leq \frac{L}{2} \|h\|^2. \quad (93)$$

Apply this at $h = \eta v$ and $h = \eta v'$. Instantaneous preservation cancels the linear terms and the triangle inequality yields Eq. (66) for the actual velocities, capped or uncapped. For active uncapped CDC,

$$\|v'\| \leq \|v\| + \frac{\delta}{\|q\|} \leq V + D/q_{\min}, \quad (94)$$

which gives the explicit bound; a nonnegative upper cap can only decrease the added multiple of q . If smoothness is known only on the fixed-coordinate ball $B(\theta, r)$, sufficient segment conditions are $\eta V \leq r$ and $\eta(V + D/q_{\min}) \leq r$. These quantities and L are chart- and metric-dependent. $\square$

C Discrete observation

For logged values Δ_k , increments telescope exactly:

$$\Delta_K - \Delta_0 = \sum_{k=0}^{K-1} (\Delta_{k+1} - \Delta_k). \quad (95)$$

For one strict block of positive increments followed by one strict block of negative increments, the peak-minus-initial value is the positive discrete area and the peak-minus-final value is the negative tail area. A discrete causal certificate requires the final gap to lie strictly below its initial value and the weak-only trajectory to have a nondegenerate first hit. Gate failure blocks the causal certificate but does not establish absence of starvation. Because these statements concern logged increments, they make no assertion between observation times; continuous crossing-time claims require the hypotheses of Corollary 14.

D Protocol details

Table 5: Experimental settings. Shadow methods train a second weak-only model and therefore use roughly twice the model memory and per-step compute.

Study	Model / width	Task grid	Samples	Steps / LR	Seeds / records	Purpose
Controlled linear grid	dense linear / 96	$\rho \in \{1, 2, 4, 8\}$, lag $\in \{0, 2, 4, 8\}$	256	500 / .01	4-11	paired deficit and gate
Finite-width reference	dense linear / 32-256	$(1, 0), (2, 2), (4, 4), (8, 8)$	1024	400 / .01	0-7	empirical reference
Nonlinear fixed cell	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	20-27	direct-drift certificate
Broad tangent pilot	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	1 $\times$ 8	five-output test
Held-out event test	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	4 $\times$ 8 / model	crossing events
Dense mechanism control	dense linear / 96	$\rho = 4$, lag 2	256	500 / .01	40-47	local CDC comparison
Generated-cue images	semi-real CNN	MNIST 3-vs-8; FashionMNIST 0-vs-6	512/256/256 per dataset	500 / .01	4 $\times$ 8 / dataset; 64	generated-cue intervention
Threshold robustness	tanh, GRU / 32	three cells; $\beta \in \{.25, .5, 1, 2\}$	256	1000 / .01	4 $\times$ 8; 192	all- β endpoint
Mechanism control	tanh / 64	two cells; ERM, CDC, adapted Bloop/PCGrad variants	256	1000 / .01	4 $\times$ 8; 256 method-records	joint tradeoff

E Additional negative and mixed results

E.1 Semi-real generated-cue study

This study contains 64 crossed records: four data seeds by eight model seeds for MNIST 3-vs-8 and the same design for FashionMNIST 0-vs-6. The core channel contains selected real benchmark pixels, and the intervention zeros only the generated cue. Positive weak-AUC gap means that weak-only core gain exceeds both-feature core gain.

Table 6: Complete semi-real generated-cue outcomes. Intervals use the crossed-seed design separately for each dataset.

Dataset	Weak-only learnable	Causal certificate	Mean weak AUC gap (95% CI)
MNIST 3-vs-8	0/32	0/32	−0.0012362842136667493 [−0.0034405428466004646, 0.0010159591371180453]
FashionMNIST 0-vs-6	0/32	0/32	−0.00010921728159018996 [−0.004129384520886106, 0.003182670049955049]

The weak-only first-hit gate fails in every record. The zero causal counts are therefore negative or indeterminate: they cannot be interpreted as evidence that starvation is absent or that either dataset is intrinsically unlearnable.

E.2 Threshold-robust extension

This extension contains 192 records: tanh and GRU across three fixed task cells and a crossed 4 $\times$ 8 seed design. Each record evaluates the first-hit grid $\beta \in \{0.25, 0.5, 1, 2\}$. The architecture estimand equally weights the three cell estimates. A 5,000-replicate two-way bootstrap supplies percentile intervals, and the two one-sided architecture tests form one Holm family.

The all- β endpoint is negative for both architectures and supports no architecture ranking.

Table 7: Complete all- β architecture endpoints.

Architecture	Macro estimate	95% CI	Holm-adjusted p
tanh	0.2916666666666667	[0.16666666666666666, 0.4166666666666667]	1.0
GRU	0	[0, 0]	1.0

E.3 Mechanism-control comparison

This comparison contains 256 method-records for ERM, CDC, and adapted Bloop-style and PCGrad-style shadow-target variants over two fixed cells and crossed seeds. Weak rescue is $-|\text{weak AUC gap}|$ and is higher-better. Trajectory and final deviations are absolute strong-response deviations from paired ERM and are lower-better.

Table 8: Complete equal-weight two-cell mechanism-control contrasts. Brackets are 95% two-way-bootstrap intervals.

Comparator	CDC-comparator weak rescue	Comparator-CDC trajectory deviation	Comparator-CDC final deviation	Joint tradeoff
Adapted Bloop-style variant	0.002198259399210656 [0.0009779187294930126, 0.003277366267916477]	3.4651361294978416 [3.3443827215131754, 3.5849715262780952]	0.6422362388111651 [0.6288088704226539, 0.6563168084365315]	true
Adapted PCGrad-style variant	-0.00015351238407674823 [-0.00047309348592838785, 0.00013248012419808214]	-0.4870988912807656 [-0.6897378944428417, -0.27379327373499074]	-0.2565436437726021 [-0.3016841153614223, -0.2087651835754514]	false

The joint weak-rescue/strong-preservation criterion is true only against the adapted Bloop-style variant and false against the adapted PCGrad-style variant. Thus CDC does not beat the latter under the joint criterion. Because both comparators are adapted response-gradient shadow-target variants, the comparison supports neither parity with the original methods nor broad CDC superiority.

F Additional tangent-surrogate results

The broad tangent pilot evaluates five outputs on 16 architecture-seed records. Correct classifications are phase 12/16, drift crossing 14/16, response crossing 14/16, causal certificate 14/16, and weak-only learnability 12/16. The joint criterion fails because tanh learnability is 4/8, below the prespecified 5/8 floor; GRU learnability is 8/8. Magnitude calibration also differs sharply by architecture: tanh response-gap, weak-response, and drift RMSE are 2.367, 2.394, and 0.486, versus 0.0163, 0.0281, and 0.0107 for GRU.

The held-out event test uses separate fresh crossed seeds. Response-crossing classification is correct in 64/64 records and drift-crossing classification in 60/64 (32/32 tanh, 28/32 GRU), with two-way-bootstrap intervals $[1, 1]$ and $[0.8125, 1]$. Other outputs are substantially weaker: phase 52/64, causal certificate 56/64, and learnability 49/64, including tanh learnability 17/32. Predictions are systematically early (overall drift/response bias $-0.822/-1.189$), while tanh response-gap, weak-response, and drift RMSE are 2.455, 2.489, and 0.490. The result is therefore an uncalibrated crossing-event classifier at one controlled cell, not phase, causal, learnability, time, magnitude, prevalence, or kernel-stability validation.

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